Ambati Swechcha

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Education

New Jersey Institute of Technology Master of Science in Computer Science — Sept 2025

- Recipient of the Graduate Provost Scholarship
- GPA: 3.83
- **Coursework:** Operating Systems Design, Database Management System Design, Python for Web API Development, AI-Assisted Software Engineering, Data Structures, Machine Learning

Mahindra University Bachelor of Technology in Computer Science and Engineering — Aug 2021 – Jun 2025

- GPA: 7.82/10
- **Relevant Coursework:** Design and Analysis of Algorithms, Operating Systems, Database Management Systems, Software Engineering, Object-Oriented Programming, Distributed Systems, Machine Learning, Statistics and Probability

Technical Skills

Languages: Python, Java, JavaScript, TypeScript, SQL, C, C++
Frontend: React, Tailwind CSS, Vite, Zustand, React Router
Backend: FastAPI, Spring Boot, REST APIs, JWT Authentication, WebSockets
Databases: PostgreSQL, MySQL
AI/ML: PyTorch, TensorFlow, NLP, Transformers, OpenAI APIs, Machine Learning
DevOps & Cloud: Docker, GitHub Actions, AWS, Linux, Maven, Flyway, Playwright, PyTest

Experience

Teaching Assistant, CS 490, New Jersey Institute of Technology — Sept 2026 – Present

- Serve as a project mentor for student software projects in CS 490.
- Hold weekly office hours to help students with course projects and assignments.

Library Student Associate (Graduate, Opening/Closing Shift), Van Houten Library, NJIT — Sept 2026 – Present

- Open and close the library on scheduled shifts and secure the building.
- Handle circulation tasks: check-ins, check-outs, renewals, and group study room reservations.
- Assist patrons in person, by phone, and by email with directions and locating library materials.
- Provide basic computer and printing support when the lab assistant is unavailable.

Summer Research Intern, QMatter Labs LLC. — Jun 2024 – Aug 2024

- Improved a hybrid quantum genetic algorithm for optimizing EV charging station placement.
- Integrated datasets and enhanced the fitness evaluation function for improved optimization accuracy.
- Developed performance analysis metrics to evaluate algorithm efficiency and scalability.

Projects

MyDay – Desktop Day Planner

- Built an offline cross-platform (Linux/Windows) desktop planner with daily energy, hydration, meal, and mood check-ins, an energy-aware Pomodoro to-do list, and a rolling 7-day week planner.
- Implemented desktop notifications and local JSON persistence via a secure Electron preload bridge, with a GitHub Actions pipeline that builds and publishes AppImage and Windows installers.
- **Stack:** Electron, React, TypeScript, Vite, Tailwind CSS, GitHub Actions

FlowBoard

- Engineered an AI-assisted collaborative project management platform featuring Kanban workflows, real-time team collaboration, and intelligent board generation from project descriptions.
- Implemented live drag-and-drop task synchronization, JWT-secured role-based access control, WebSocket-based updates, and AI-powered meeting transcript summarization with action-item extraction.
- **Stack:** React, TypeScript, Spring Boot, PostgreSQL, WebSockets, JWT Authentication, Docker, Tailwind CSS, AWS

AI-Powered Lesson Generator

- Built a full-stack platform that converts topic descriptions into structured programming curricula with automated tests runnable via GitHub Actions.
- Implemented FastAPI backend, React frontend, and AI integration using OpenAI GPT-4/3.5 with fallback mechanisms.
- **Stack:** Python, FastAPI, React, TypeScript, SQLAlchemy, Docker, GitHub Actions, PyTest

BREAD Calculator with JWT Authentication and CI/CD

- Built a FastAPI application with JWT-based authentication and complete BREAD functionality.
- Implemented automated testing and CI pipelines using GitHub Actions and Playwright.
- **Stack:** Python, FastAPI, PostgreSQL, JWT, Docker, GitHub Actions

Breast Cancer Classification using Multi-Scale Deep Equilibrium Model

- Implemented a Deep Equilibrium Model for multi-scale mammogram image classification with integrated explainable AI techniques.
- **Stack:** Python, PyTorch, TensorFlow

Cricket Commentary Analysis using NLP

- Analyzed cricket commentary transcripts to identify key overs and generate summaries using NLP and sentiment analysis.
- **Stack:** Python, BERT, BART, NLTK, Word2Vec, PyTorch

Research Lab Management System

- Developed a modular Python-based system for managing research lab equipment, members, projects, and reports.
- **Stack:** Python, PostgreSQL

Leadership & Certifications

- President of Qubit, the Quantum Computing Club (2023 – 2024)
- Completed IBM Qiskit Quantum Explorers program and received Advanced Level Badge
- Completed coursework in Quantum Computing and Quantum Machine Learning through IBM Qiskit and The Coding School